

$$T(1) = 1, \quad n = 1$$

S: tratta di divide and conquer, dove abbiamo la divisione in $a=7$ problemi di dimensione $n/2$ e un costo di ricombinazione pari a n^2 .

$$T(n) = 7T(n/2) + n^2$$

$$= 7^4 T\left(\frac{n}{2^4}\right) + 7^3 \cdot \left(\frac{n}{2^3}\right)^2 + 7^2 \left(\frac{n}{2^2}\right)^2 + 7^1 \left(\frac{n}{2^1}\right)^2 + 7^0 \left(\frac{n}{2^0}\right)^2 =$$

Perciò, in generale, posso scrivere che:

$$T(n) = 7^{\log_2 n} \cdot 1 + n^2 \sum_{i=0}^{\log_2 n - 1} \left(\frac{7}{4}\right)^i = n^{\log_2 7} + n^2 \left(\frac{\left(\frac{7}{4}\right)^{\log_2 n} - 1}{\frac{7}{4} - 1} \right) = n^{\log_2 7} + \frac{1}{3} n^2 \left(\frac{n^{\log_2 7}}{n^2} - 1 \right) =$$

$$= n^{\log_2 7} \left(1 + \frac{4}{3}\right) - \frac{4}{3} n^2 = \frac{7}{3} n^{\log_2 7} - \frac{4}{3} n^2 \Rightarrow O(n^{\log_2 7})$$

$$\rightarrow \log_2 7 = \frac{\ln 7}{\ln 2} \approx 2.80$$

A/0, C/12, D/9, F/31, *, *, H/8, *, *, Z/54, Q/19, R/61, N/7, P/21, *

Diagram illustrating a game tree for a 3-player game. The root node is labeled $A/0$. The tree branches into several nodes, each labeled with a player's move and a payoff vector. The nodes are:

- $A/0$ (Root)
- $C/12$ (Branch from $A/0$)
- $D/9$ (Branch from $C/12$)
- $F/31$ (Branch from $D/9$)
- $C/12$ (Branch from $F/31$)
- $D/9$ (Branch from $C/12$)
- $A/0$ (Branch from $D/9$)
- $D/9$ (Branch from $A/0$)
- $H/8$ (Branch from $D/9$)
- $A/0$ (Branch from $H/8$)
- $A/0$ (Branch from $A/0$)
- $Z/154$ (Branch from $A/0$)
- $Z/154$ (Branch from $Z/154$)
- $A/0$ (Branch from $Z/154$)
- $Q/19$ (Branch from $Z/154$)
- $R/61$ (Branch from $Q/19$)
- $Z/154$ (Branch from $R/61$)
- $Q/19$ (Branch from $R/61$)
- $A/0$ (Branch from $Z/154$)
- $N/7$ (Branch from $A/0$)
- $R/61$ (Branch from $N/7$)
- $Z/154$ (Branch from $R/61$)
- $Q/19$ (Branch from $R/61$)
- $A/0$ (Branch from $Z/154$)
- $N/7$ (Branch from $A/0$)
- $Q/19$ (Branch from $N/7$)
- $Z/154$ (Branch from $Q/19$)
- $P/121$ (Branch from $Z/154$)
- $Q/19$ (Branch from $P/121$)
- $A/0$ (Branch from $Q/19$)
- $N/7$ (Branch from $A/0$)

Es. 3) LIS 7, 4, 6, 3, 8, 9, 2, 10, 5

SVOLGIMENTO

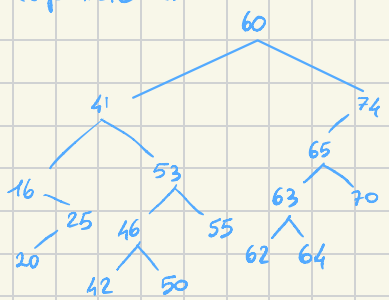
Start:

	0	1	2	3	4	5	6	7	8
	7	4	6	3	8	9	2	10	5
P	-1	-1	-1	-1	-1	-1	-1	-1	-1
L	1	1	1	1	1	1	1	1	1

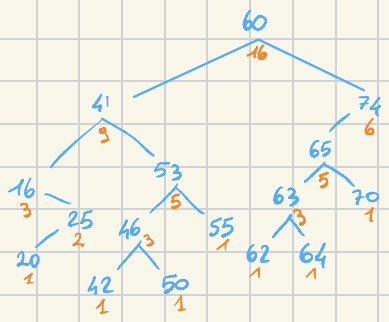
	0	1	2	3	4	5	6	7	8
	7	4	6	3	8	9	2	10	5
P	-1	-1	1	-1	2	4	-1	5	1
L	1	1	2	1	3	4	1	5	2

→ LIS: 4 6 8 9 10

Es. 4) Trasformare in os-BST e ricerca chiave rango 6:

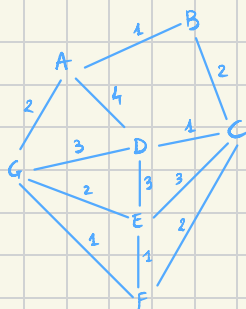


SVOLGIMENTO:



$6 < 3+1 \quad 60 \rightarrow 41$
 $6 > 3+1 \quad 41 \rightarrow 53$
 $2 < 3+1 \quad 53 \rightarrow 46$
 $2 = 1+1 \quad \Rightarrow 46$
 $K = 6 - 3 - 1 = 2$

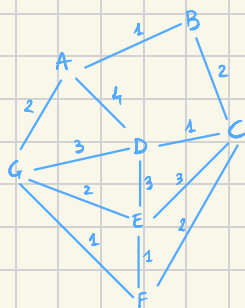
Es. 5) Kruskal



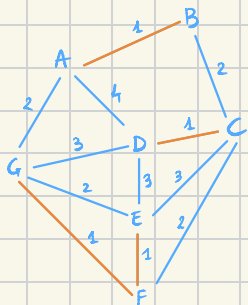
SVOLGIMENTO

Archi:

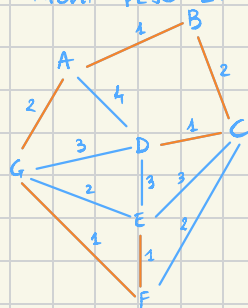
- peso 1: A-B, C-D, E-F, F-G
- peso 2: A-G, B-C, C-F, E-G
- peso 3: C-E, D-E, D-G
- peso 4: A-D



- Archi peso 1:



- Archi peso 2:

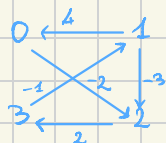


Mi fermo a B-C.

Tot = 8

Es. 6) Bellman-Ford da 1.

SVOLGIMENTO



ordine archi:

$0 \rightarrow 2, 1 \rightarrow 0, 1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 1$

PASSI

	0	1	2	3	4	...
0	∞	4	2	0	-2	
1	0	∞	-4	-6	-8	
2	∞	-3	∞	-7	-5	
3	∞	-1	-3	∞	-7	

È stato rilevato un ciclo a peso negativo:

$1 \rightarrow 2 \rightarrow 3 \rightarrow 1: -2$

Quindi non è possibile raggiungere un punto fisso.
Punto fisso: ulteriori rilassamenti degli archi non migliorano le stime dei cammini minimi.